JAN-G-338

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JOINT ARMY-NAVY SPECIFICATION

GLUE, ANIMAL

(For Use in the Manufacture of Ammunition Primers)

Army Number	Navy Number
50-11-112A	52G14

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
50-11-112	Bureau of Ordnance
4 Nov. 1943	O.S. 1345
	2 Sept. 1942

A. APPLICABLE SPECIFICATIONS

A-1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

U. S. ARMY SPECIFICATION

100-2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

B. GRADE

B-1. This specification covers one grade of animal glue as hereinafter specified.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—Animal glue shall be manufactured from raw materials of animal origin.

C-2. *Workmanship.*—Animal glue shall be manufactured by such processes as to insure compliance with the requirements hereinafter specified and shall be free from visible impurities.

D. GENERAL REQUIREMENTS

D-1. See section E.

E. DETAIL REQUIREMENTS

E-1. *Form.*—The glue shall be either regular ground or in the form of chips or flakes of a thickness not greater than ¼ inch.

E-2. *Odor.*—There shall be no odor of decomposition after a solution of the glue is heated as described in paragraph F-4a.

E-3. *Moisture.*—The moisture content of the sample when tested in accordance with paragraph F-4b shall be not less than 9 percent and not more than 15 percent.

E-4. *Jelly strength.*—The glue shall show a jelly strength of not less than 170 gm and not more than 194 gm.

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[1] Applicable only to Navy purchases.

E–5. *Ash.*—The ash content shall not exceed a maximum of 4 percent.

E–6. *Adulterants.*—When tested in accordance with paragraph F–4e the glue shall show no evidence of the presence of alum, zinc oxide, salts, sugar, glucose, dextrin, or clay fillers.

E–7. *Neutrality.*—When tested in accordance with paragraph F–4f, the glue shall exhibit a pH within the range of 6.5 to 7.5.

E–8. *Sulfides and sulfites.*—When tested in accordance with paragraph F–4g the glue shall show no evidence of the presence of sulfides or sulfites.

E–9. *Grease.*—The grease content shall not exceed a maximum of 0.4 percent.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—A lot shall consist of not more than 1000 pounds.

F–2. *Sampling.*—A minimum of 10 percent of the containers shall be selected by the Government inspector so as to be representative of the lot. When lots comprise less than 100 containers, either 10 containers or all containers shall be taken. Approximately 4 ounces of material shall be removed from each selected container. The primary samples shall be thoroughly mixed and reduced to such a degree of fineness that the composite sample will pass through a U. S. Standard No. 10 sieve. Quarter the sample until a composite sample of approximately 1 pound is obtained. The composite sample shall be placed in an airtight container and labeled to show the name of the material, manufacturer, plant, contract or order number, lot number and number of pounds in the lot. All acceptance tests shall be made on the composite sample representative of the lot.

F–3. *Inspection.*—Inspection shall be at the point of delivery unless otherwise specified in the contract or order. The material shall be visibly inspected during sampling to determine the absence of foreign matter, lumps and large particles (see par. E–1).

F–4. *Tests.*—The laboratory tests shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Odor.*—Transfer approximately 50 mℓ of a 10 percent solution of the glue, or a portion of the solution used in the jelly strength test, to an uncovered container and heat in an incubator at 37° to 38° C. for 48 hours. Examine the solution for odor of decomposition.

F–4b. *Moisture.*—Transfer an accurately weighed portion of approximately 10 gm of the sample to a tared porcelain dish and dry in a well ventilated oven at 105° ± 3° C. for 16 hours. Cool in a desiccator and weigh. Calculate the loss in weight as precentage of moisture in the sample. Retain the residue for the ash determination (see par. F–4d).

F–4c. *Jelly strength.*—

F–4c(1). *Preparation of sample.*—Prepare solutions of the glue in triplicate. Transfer sufficient of the glue sample (calculated from the moisture content previously determined as specified in par. F–4b) to give 13.2 gm of dry glue (15 gm at 12 percent moisture content) to a standard 150 mℓ wide-mouthed bottle having an inside diameter of 59 mm, outside diameter of 66 mm, overall height of 85 mm and fitted with a No. 9 rubber stopper. Add enough distilled water to the glue to make a total of 106.8 gm of water, including that naturally contained in the glue (105 gm if the glue has 12 percent moisture content). This will make an approximately 11 percent solution of anhydrous glue. Mix the glue and water by stirring with a glass rod and allow to stand at a temperature of 10° to 15° C. for at least 4 hours. Melt the glue by heating the samples to a temperature of 62° C. in a water bath in which the temperature of the water shall not exceed 70° C. The time required to bring the samples to 62° C. in a water bath in which the temperature of the water shall not exceed 70° C. The time required to bring the samples to

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exceed 15 minutes. The glue solution shall be free of air or foam. Allow the samples to cool in the standard stoppered bottles to approximately 45° C. and mix thoroughly with the solution any water of condensation. Place the samples in a chamber or bath controlled at 10° ±0.1° C. for 16 to 18 hours.

F–4c(2). *Determination.*—Measure the jelly strength on at least 3 separate quantities of the solutions prepared and bottled in accordance with paragraph F-4c(1), and average the results. Determine the jelly strength on a Bloom gelometer adjusted to give a 4 mm depression and to deliver a load of 40 gm per second to the surface of the jelly. The test must be completed within 2 minutes after the removal of the sample from the cooling chamber.

F–4d. *Ash.*—Ignite the residue from the moisture determination (see par. F–4b) to constant weight at a red heat. Care should be taken to ignite gently, at first, and to increase the temperature gradually. Cool the dish in a desiccator and weigh. Calculate the weight of residue as percentage of ash in the sample. Retain the ash for the determination of adulterants (see par. F–4e(1)).

F–4e. *Adulterants.*—

F–4e(1). *Alum and zinc oxide.*—Dissolve the residue from the ash determination (see par. F–4d) in hydrochloric acid and qualitatively test for the presence of alum and zinc oxide.

F–4e(2). *Sugar, glucose, and dextrin.*—Transfer approximately 0.2 gm of the glue to a test tube, add 10 mℓ of water containing a few drops of concentrated hydrochloric acid. Heat in boiling water for 5 minutes. Add 5 mℓ of Benedict's solution, mix, and continue heating for 2 or 3 minutes. The appearance of a turbidity indicates the presence of sugar, glucose, or dextrin.

F–4f. *Neutrality.*—Determine the pH of the glue colorimetrically or electrometrically, at 60° C. Use portions of the solutions upon which the jelly strength tests were made, or other samples of equal concentration. If pH is determined colorimetrically mix 15 mℓ of the glue solution at 60° C. with 20 mℓ of distilled water at the same temperature. Place 10 mℓ portions of the diluted glue solution in test tubes. To one portion add 6 to 8 drops of a 0.02 percent solution of methyl red. To the other add 6 to 8 drops of a 0.04 percent solution of bromthymol blue, as indicators. Determine the pH by comparing the color of the test solutions with a set of standard colors.

F–4g. *Sulfides and sulfites.*—Add chemically pure hydrochloric acid to a portion of the softened glue. Any odor of H_2S or SO_2 shall be cause for rejection.

F–4h. *Grease.*—Determine the grease content by the Kissling method [1] or other satisfactory extraction method which indicates total grease.

F–5. *Retest.*—If a lot of glue fails on inspection or test, no retest will be permitted.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—The glue shall be packed in wooden boxes or kegs of uniform dimensions, and the net weight of glue in a box or keg shall not exceed 100 pounds. All containers shall be in accordance with the applicable regulations of the Interstate Commerce Commission and shall be so constructed as to insure acceptance of material by common or other carriers for safe transportation at the lowest rate to the point of delivery.

[1] The Kissling and other extraction methods of determining total grease content are described in Fernback's "Glue and Gelatine" (1907) and Bogue's "Chemistry and Technology of Gelatine and Glue" (1922).

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G–2. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES

H–1. Requests, requisitions, schedules, and contracts or orders should specify the title of the specification, the number, and the date.

H–2. *Use.*—Animal glue covered by this specification is intended for use as a raw material in the manufacture of primers.

H–3. Copies of Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department". Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, and U. S. Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–4. Copies of Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–5. Copies of this Joint Army-Navy specification required for Army purchases may be obtained upon application to the Office, Chief of Ordnance, Army Service Forces, Washington 25, D. C.

H–6. Copies of this Joint Army-Navy specification required for Navy purchases may be obtained as indicated in paragraph H–4.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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